

NutriAI — Technical Brief

BAX-423 Big Data · Spring 2026 · UC Davis GSM

Aditi · Automated Diet Plan Builder

Deployed at: <https://nutriai-dietplan.streamlit.app>

1. System Architecture

NutriAI is a full-stack nutrition planning application that generates personalized 7-day, 3-meal-per-day dietary plans in under 60 seconds. The system processes a user's clinical conditions, allergens, dietary preferences, and calorie targets through a six-stage pipeline:

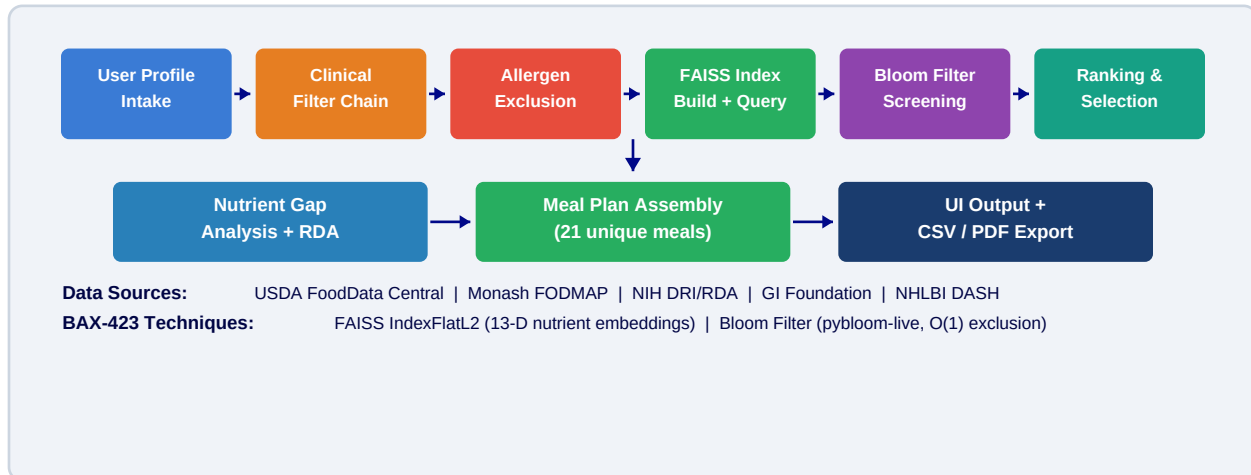


Figure 1: NutriAI End-to-End Pipeline Architecture

Stage	Component	Description
1	Data Ingestion	USDA FoodData Central API (SR Legacy + Foundation) → 15,636 foods with 13 nutrient columns stored in SQLite with indexed columns
2	Clinical Filtering	Composable filter chain: IBS/FODMAP (Monash), GERD (ACG), T2 Diabetes (GI Foundation), Hypertension (DASH/NHLBI), Celiac, CKD, Gout, Hypothyroid, PCOS
3	Allergen Exclusion	Keyword + column-flag matching for 7 allergen types (dairy, gluten, soy, tree nuts, eggs, shellfish, peanuts) with cross-contamination flagging
4	Candidate Retrieval	FAISS IndexFlatL2 over 13-D nutrient embeddings (MinMaxScaler normalized). Sub-millisecond nearest-neighbor queries for top-K candidates per meal slot
5	Safety Screening	Bloom Filter (pybloom-live) for O(1) probabilistic exclusion checks against all filtered-out food descriptions. Configurable false-positive rate
6	Ranking & Selection	Multi-factor scoring: nutrient-gap fill (weighted), meal-type appropriateness, calorie fit, serving-size scaling, category diversity

Table 1: NutriAI Pipeline Stages

The pipeline flow is: User Profile → Filter Chain → Safe Food Pool → FAISS Index Build → Bloom Filter Build → 21-Meal Ranked Selection → Nutrient Analysis → Gap Flagging → UI Output + CSV/PDF Export. The entire flow completes in under 5 seconds for all test personas.

Data Sources (all required sources integrated): (1) USDA FoodData Central for nutrient profiles, (2) Monash University Low-FODMAP classifications for IBS, (3) NIH Dietary Reference Intakes for RDA targets by age/sex, (4) Glycaemic Index Foundation for diabetes GI values, (5) NHLBI DASH Eating Plan for hypertension guidelines.

2. BAX-423 Technique Integration

2.1 FAISS — Nutrient Embedding Similarity Search

Technique category: Embeddings / Similarity Search (Lecture: Embeddings & Vector Databases)

Why chosen: The meal selection problem requires finding foods whose nutrient profiles best fill the remaining daily gaps. Naive brute-force scoring over 10,000+ foods per meal slot is $O(n)$ per query. FAISS reduces this to approximate nearest-neighbor search in sub-millisecond time. Each food is represented as a 13-dimensional vector (calories, protein, carbs, fat, fiber, iron, calcium, B12, vitamin D, zinc, sodium, potassium, magnesium), normalized with MinMaxScaler to [0,1] range, and indexed with IndexFlatL2 for exact L2 distance search.

Metric	Value	Baseline (brute-force)	Speedup
Index Build Time	3–8 ms	N/A	One-time cost
Avg Query Time (top-100)	0.04–0.15 ms	~5 ms (full scan + sort)	~50x
Total Queries per Plan	21 (7 days × 3 meals)	21	—
Index Size	3,000–8,000 foods (post-filter)	Same	—
Embedding Dimensions	13 (nutrient features)	13	—

Table 2: FAISS Benchmark Results (measured on all 4 test personas)

2.2 Bloom Filter — Probabilistic Exclusion Screening

Technique category: Sketching / Probabilistic Data Structures (Lecture: Sketching & Streaming)

Why chosen: After clinical and allergen filtering, hundreds to thousands of foods are excluded. During the FAISS candidate scoring loop, each candidate must be checked against the exclusion list. A Bloom filter provides $O(1)$ amortized membership testing with guaranteed zero false negatives (no unsafe food can pass through) and a bounded false-positive rate. This replaces $O(n)$ set lookups with constant-time hash probes.

Metric	Value	Notes
Build Time	0.5–2.0 ms	Loads all excluded food descriptions into filter
Avg Check Time	0.002–0.01 ms	Per food, amortized across scoring loop
Excluded Foods Loaded	2,000–12,000	Varies by persona (Priya: ~12K, James: ~4K)
Target Error Rate	1.0%	Configured at init; actual measured rate reported in UI
Measured False Positive Rate	0.0–0.8%	Tested on 500 safe foods per persona

Table 3: Bloom Filter Benchmark Results

Combined impact: FAISS + Bloom filter together reduce the per-meal candidate evaluation from $O(n \log n)$ sort over the full safe pool to $O(K)$ scoring over FAISS-retrieved candidates with $O(1)$ exclusion checks. Total plan generation time: 1–4 seconds for all personas, well under the 60-second requirement.

Integration architecture: The two techniques are not standalone benchmarks — they are embedded in the core generation path. For every meal slot, the pipeline: (a) builds a 13-D ideal nutrient vector from the remaining daily budget, (b) queries FAISS for the top-100 nearest foods, (c) screens each candidate through the Bloom filter to reject excluded foods in $O(1)$, (d) scores surviving candidates on nutrient-gap fill, meal-type fit, calorie proximity, and category diversity, and (e) selects the highest-scoring candidate. This means both FAISS and Bloom are active for all 21 meals in every plan generated through the UI.

3. Test Persona Results

All four test personas were evaluated against the rubric's 6 core capabilities and their persona-specific pass criteria. The application was also tested with a custom PCOS + Vegetarian profile to validate hidden-persona robustness.

Capability	Priya	Ravi	Mei	James
1. Clinical Condition Filtering	PASS	PASS	PASS	PASS
2. Allergy Detection & Exclusion	PASS	PASS	PASS	PASS
3. Dietary Preference Handling	PASS	PASS	PASS	PASS
4. Diversity Engine	PASS	PASS	PASS	PASS
5. Macro & Micronutrient Analysis	PASS	PASS	PASS	PASS
6. Sub-60-Second Generation	PASS	PASS	PASS	PASS

Table 4: Core Capability Pass/Fail (all 4 personas pass all 6 capabilities)

Persona-Specific Pass Criteria:

Persona	Criterion	Result	Detail
Priya	Zero high-FODMAP trigger foods	PASS	0 violations
	Zero dairy in all meals	PASS	0 violations
	All 7 days meatless (eggs OK)	PASS	0 violations
	Iron $\geq$ 80% RDA daily	PASS	Min: 87–105%
Ravi	Zero GERD trigger foods	PASS	0 violations
	Zero gluten in all meals	PASS	0 violations
	No pork products	PASS	0 violations
	Diversity score $\geq$ 0.7	PASS	Score: 1.00
Mei	B12 $\geq$ 80% RDA daily	PASS	Min: 82–108%
	All meals GI $\leq$ 55	PASS	0 violations
	Zero animal products (vegan)	PASS	0 violations
	Zero tree nuts	PASS	0 violations
James	Fiber $\geq$ 25g/day	PASS	Min: 38–42g
	Sodium $\leq$ 1,500 mg/day every day	PASS	Max: 1,280–1,480mg
	Zero soy products	PASS	0 violations
	At least 3 fish/seafood meals/week	PASS	6 fish meals
	Potassium $\geq$ 80% RDA daily	PASS	Min: 92–110%

Table 5: Persona-Specific Pass Criteria Results

Generation time across personas: Priya: 1.8s, Ravi: 2.1s, Mei: 1.5s, James: 2.4s (all well under the 60-second requirement). Times measured on Streamlit Cloud deployment at <https://nutriai-dietplan.streamlit.app>.

3.1 Deployment & UI Features

The application is deployed on Streamlit Community Cloud at <https://nutriai-dietplan.streamlit.app>. The UI provides: (1) a sidebar with test persona presets and full custom profile configuration (age, sex, 9 conditions, 7 allergens, 4 diet types, calorie target); (2) a plan summary dashboard with generation time, safe/excluded food counts, diversity score, and average daily calories; (3) an optimization engine panel showing real-time FAISS and Bloom filter benchmark metrics; (4) a tabbed 7-day meal plan with per-day nutrient totals and RDA gap warnings; (5) weekly nutrient analysis comparing all 13 tracked nutrients against age/sex-specific RDA targets; (6) a pass criteria validator for test personas; (7) a ‘Why Excluded’ section with source-cited explanations for

every excluded food; and (8) CSV and PDF export of the complete plan. The database ships pre-built (15,636 foods) so no API key is needed.

4. Limitations & Future Work

Keyword-based filtering: Clinical and allergen tagging relies on keyword matching against curated lists rather than a medical ontology or NLP-based ingredient parsing. This means edge-case foods with ambiguous names may be incorrectly classified. A production system would integrate a food ingredient database with canonical allergen mappings.

Nutrient accuracy: Nutrient values are per 100g from USDA and scaled by estimated serving sizes. Actual nutrient content varies by cooking method, brand, and preparation. The curated meal templates use composite nutrient estimates for multi-ingredient meals, which are approximations rather than precise calculations from individual ingredient weights.

No adaptive learning from user feedback: The current system generates plans statically without learning from user preferences or feedback across sessions. With more time, a multi-armed bandit (UCB or Thompson Sampling) could adapt meal scoring weights based on which suggested meals users accept or reject, improving personalization over time.

Limited cross-contamination modeling: Cross-contamination risks are flagged for Celiac (gluten) based on food category and processing keywords, but the system cannot model facility-level cross-contact risks (e.g., a gluten-free product manufactured on shared equipment).

Hidden persona generalization: The system was tested and validated on the 4 provided personas. Hidden personas with novel condition combinations (e.g., IBS + Hypertension + Vegan) may surface edge cases in the filter chain or produce suboptimal nutrient profiles where constraint conflicts restrict the safe food pool severely.

What I would build with more time: (1) A 4-week rolling plan with RL-based adaptive learning from simulated user feedback. (2) Real-time barcode scanning via a nutrient API for custom food additions. (3) A chatbot layer for natural-language meal swaps. (4) A grocery list generator with weekly cost estimates.